

1 — Regression / Data Fitting (Th.1–6)

Regression ≠ interpolation: interpolation trusts every measurement and passes through ALL points; regression assumes noise and finds the best OVERALL trend, passing through (almost) none. Deviations $d_i = y_i - \phi(x_i)$, data (x_i, y_i) , $i = 0..n$ ($n+1$ points). **Error choices:** naive $S = \sum d_i$ (rejected: +, − cancel); abs $S = \sum |d_i|$ (rejected: kink at 0, no derivative); **squared** $S = \sum d_i^2$ (chosen: smooth everywhere, so “set gradient = 0” gives solvable linear equations; big misses punished quadratically). Example (Lec14): pts (1, 2), (2, 1), line through (1, 1), (2, 2): $d_0 = 1, d_1 = -1$: naive $S = 0$ (cancels!), abs $S = 2$, squared $S = 2$.

Linear regression

$\hat{y} = a_0 + a_1x$, $S = \sum (y_i - a_0 - a_1x_i)^2$. General fit $\phi(x, a_0..a_\ell)$: $\frac{\partial S}{\partial a_k} = -2 \sum_{i=0}^n (y_i - \phi(\dots)) \frac{\partial \phi}{\partial a_k} = 0$, $k = 0..l$ ($\ell+1$ equations). For the line this gives the two normal equations below. ∂_{a_0} : $\sum y_i - (n+1)a_0 - a_1 \sum x_i = 0$; ∂_{a_1} : $\sum x_i y_i - a_0 \sum x_i - a_1 \sum x_i^2 = 0$. Divide by $n+1 \Rightarrow a_0 + a_1 \bar{x} = \bar{y}$, $a_0 \bar{x} + a_1 \bar{x}^2 = \bar{x}\bar{y}$; Cramer gives

$$a_1 = \frac{\bar{xy} - \bar{x}\bar{y}}{\bar{x}^2 - \bar{x} \cdot \bar{x}}, \quad a_0 = \bar{y} - a_1 \bar{x}$$

with $\bar{x} = \frac{1}{n+1} \sum x_i$, $\bar{x}^2 = \frac{1}{n+1} \sum x_i^2$, $\bar{xy} = \frac{1}{n+1} \sum x_i y_i$.

Trap: $\bar{x}^2 \neq \bar{x}^2$; denominator = 0 iff all x_i equal (no fit possible).

Hessian + minimum test

$$H = \begin{pmatrix} S_{a_0 a_0} & S_{a_0 a_1} \\ S_{a_1 a_0} & S_{a_1 a_1} \end{pmatrix}$$

$$S_{a_0 a_0} = 2(n+1), \quad S_{a_0 a_1} = 2 \sum x_i, \quad S_{a_1 a_1} = 2 \sum x_i^2$$

$$\det H = 4 \left[(n+1) \sum x_i^2 - \left(\sum x_i \right)^2 \right]; \min \iff S_{a_0 a_0} > 0 \text{ AND } \det H > 0$$

Intuition: H = curvature of the error bowl; positive-definite = bowl opens upward = exactly one bottom. **Cauchy–Schwarz** $\Rightarrow \det H \geq 0$; > 0 if ≥ 2 distinct $x_i \Rightarrow$ unique global min (strictly convex).

Multivariable + pseudoinverse

$\hat{y} = a_0 + a_1x_1 + \dots + a_mx_m$, X is $(n+1) \times (m+1)$ with first col of 1's. $S = \|\bar{y} - X\bar{a}\|^2$.

$$X^T X \bar{a} = X^T \bar{y}, \quad \bar{a} = \underbrace{(X^T X)^{-1} X^T \bar{y}}_{X^+}$$

$X^+ =$ Moore–Penrose pseudoinverse ($= X^{-1}$ when X square). Idea: $X^T X \bar{a} = X^T \bar{y}$ projects $\bar{y}$ onto the column space of X (closest achievable fit); X^+ is that projection written as a formula. **Trap:** $X^+ X = I$ holds; $XX^+ = I$ generally does NOT.

ML: classification

Features = measured inputs (x_1, x_2) ; model outputs a score; plane $a_0 + a_1x_1 + a_2x_2 = 0 \cap x_1x_2$ -plane = **decision boundary** (class 1 vs 2). Logistic $\sigma(z) = \frac{1}{1+e^{-z}} : \mathbb{R} \rightarrow [0, 1]$ squashes the score to a probability; classifier $\sigma(a_0 + a_1x_1 + a_2x_2)$.

2 — Linearization

Goal: transform $(\tilde{x}, \tilde{y})$ so model is linear; fit; inverse-transform back. Why logs work: $\ln$ turns $\times$ into $+$ and powers into $\times$ (e.g. $\ln(Ae^{Bx}) = \ln A + Bx$). Steps: (1) change vars; (2) fit $\tilde{y} = a_0 + a_1\tilde{x}$; (3) recover A, B ; (4) check fit.

- $y = Ae^{Bx}$: $\tilde{x} = x$, $\tilde{y} = \ln y$; $a_0 = \ln A$, $a_1 = B \iff A = e^{a_0}$.
- $y = Ax^n$: $\tilde{x} = \ln x$, $\tilde{y} = \ln y$; $a_0 = \ln A$, $a_1 = n$.
- $y = A + \frac{B}{x}$: $\tilde{x} = \frac{1}{x}$, $\tilde{y} = y$; $a_0 = A$, $a_1 = B$.
- $y = \frac{1}{A+Bx}$: $\tilde{x} = x$, $\tilde{y} = \frac{1}{y}$; $a_0 = A$, $a_1 = B$.

$$\bullet \quad y = \frac{Ax}{B+x}: \tilde{x} = \frac{1}{x}, \tilde{y} = \frac{1}{y}; a_0 = \frac{1}{A}, a_1 = \frac{B}{A} \iff A = \frac{1}{a_0}, B = \frac{a_1}{a_0}.$$

Trap: not everything linearizes (e.g. $a_0e^{-x} \sin(a_1 + 2\pi x)$). **Model from graph:** straight $\rightarrow$ linear; explosive growth $\rightarrow Ae^{Bx}$; power curve $\rightarrow Ax^n$; saturating hyperbola $\rightarrow Ax/(B+x)$; reciprocal decay $\rightarrow A+B/x$.

3 — Grids + Riemann Sums

Partition $a = x_0 < \dots < x_n = b$, $h_i = x_i - x_{i-1}$, sample $x_i^* \in [x_{i-1}, x_i]$: $\int_a^b f \approx \sum f(x_i^*)h_i$.

Uniform: $h_i \equiv h = \frac{b-a}{n}$, $n+1$ points.

Non-uniform: varying h_i (adaptive: fine where f changes fast, coarse elsewhere). Idea: spend grid points where the function wiggles; a uniform grid wastes them where f is flat and starves them where f is steep. **Riemann integral** $= \lim_{\max h_j \rightarrow 0} \sum h_i f(x_i^*)$; f bounded, at most countably many jumps (isolated jumps are OK, the area under one point is zero).

4 — Rectangle / Trapezoid / Simpson

Sample point: $x_i^* = \alpha x_{i-1} + (1-\alpha)x_i$; $\alpha = 1$ left, $\alpha = 0$ right, $\alpha = \frac{1}{2}$ midpoint. (Quiz7: $0.2x_{i-1} + 0.8x_i$ is $\alpha = 0.2$; Lec16: $x_{i-1} + h_i/3$ is $\alpha = \frac{2}{3}$, $x_{i-1} + 2h_i/3$ is $\alpha = \frac{1}{3}$.) General non-uniform: left $\sum h_i f(x_{i-1})$, right $\sum h_i f(x_i)$, mid $\sum h_i f(\frac{x_{i-1}+x_i}{2})$.

$$\tilde{I}_{\text{Tr}} = \frac{1}{2} \sum_{i=1}^n (f(x_{i-1}) + f(x_i)) h_i$$

Uniform trapezoid: $\frac{h(f_0+f_n)}{2} + h \sum_{i=1}^{n-1} f_i$. **Trap idea:** one strip = rectangle hQ_1 + triangle $\frac{h}{2}(Q_2 - Q_1) = h(Q_1 + Q_2)/2$; uniform version averages neighbors (interior f_i counted twice $\rightarrow$ single sum). **Simpson** ($n = 2m$ even, uniform): fits parabola on each pair. Why $O(h^4)$? A parabola matches f 's value, slope AND curvature, so the leftover is two orders smaller than trapezoid (which matches only value + slope). Cost: more f -evaluations per step.

$$I_{\text{parab}} = \frac{h}{3} \left\{ f_0 + f_n + 4 \sum_{\text{odd } i} f_i + 2 \sum_{\text{even } i < n} f_i \right\}$$

Sketch: rectangles = bars at x_i^* ; trapezoids = straight segments node-to-node; Simpson = parabola arcs per pair (blank axes: mark $n+1$ nodes first). Simpson weights: endpoints $\times 1$, odd indices $\times 4$, even interior $\times 2$.

Accuracy estimates

- left/right rectangle $O(h)$; midpoint $O(h^2)$; trapezoid $O(h^2)$; Simpson $O(h^4)$.
- Read $O(h^p)$ as: halving h divides the error by 2^p (rect $\div 2$, mid/trap $\div 4$, Simpson $\div 16$).
- If $|h| < 1$: midpoint beats left/right; trap $\approx$ midpoint order.
- Left-rect global: $|E| \leq \frac{M_1(b-a)}{2} h$, $M_1 = \max |f'|$. Midpoint: $|E| \leq \frac{M_2(b-a)}{24} h^2$. (Taylor + symmetry kills f' term.)
- No elementary antiderivative ($\int e^{x^2} dx$, $\int \frac{dx}{\ln x}$): cross-check Simpson vs trapezoid.

5 — ODEs: IVP + Numeric Methods

$y' = f(x, y)$: slope field tells the direction at every point; a solution follows the arrows. General solution $y(x, C)$ = whole family of curves; IVP (Cauchy) $+ y(x_0) = y_0$ picks the ONE curve through (x_0, y_0) (fix C by plugging IC; existence/uniqueness assumed). **Check-solution trap:** differentiate candidate AND plug into ODE — both must hold for ALL C . **Bridge (Lec19):** Euler comes from the finite difference $y' \approx \frac{y_{i+1} - y_i}{h}$: replace y' in $y' = f(x, y)$ and solve for y_{i+1} .

$$y_{i+1} = y_i + hf(x_i, y_i), \quad x_{i+1} = x_i + h \quad (\text{Euler})$$

$$y_{i+1} = y_i + hf(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}}), \quad x_{i+\frac{1}{2}} = x_i + \frac{h}{2}, \quad y_{i+\frac{1}{2}} = y_i + \frac{h}{2} f(x_i, y_i)$$

RK4: weighted average of 4 trial slopes with weights (1, 2, 2, 1)/6: $k_1 = f(x_n, y_n)$; $k_2 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$; $k_3 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2)$; $k_4 = f(x_n + h, y_n + hk_3)$. Price: 4 f -evaluations per step for $O(h^4)$. **Global errors:** Euler $O(h)$ (local $O(h^2) \times N \sim 1/h$); midpoint $O(h^2)$; RK4 $O(h^4)$ (4 evals/step). **Euler error idea:** Taylor $y(x+h) = y(x) + hy'(x) + \frac{h^2}{2}y''(\xi)$; one Euler step from exact $y(x_i)$ misses by $O(h^2)$ (local); $N = (b - x_0)/h = O(1/h)$ steps accumulate to $O(h)$ global. **Midpoint idea:** predict midpoint slope with one Euler half-step, then step full h along it; symmetric $\Rightarrow$ local $O(h^3)$, global $O(h^2)$.

6 — Linear Algebra* (starred)

det = volume scale factor of the matrix map; det $\neq 0 \iff$ invertible (det = 0 = space collapses). $\det(AB) = \det A \det B$. $Ax = b$ has a unique solution $\iff \det A \neq 0 \iff n$ pivots (else 0 or ∞ solutions). **Basic ops:** $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{pmatrix} = \begin{pmatrix} 17 \\ 39 \end{pmatrix}$ (row $\cdot$ column).

Laplace: expand along row/col with $(-1)^{i+j}$ cofactors (pick the row/col with most zeros). Elimination: row ops change det by sign/scale (swap $\rightarrow -$, scale row by $c \rightarrow \times c$); det = product of U diagonal.

LU: $A = LU$ records elimination (L = unit-lower multipliers, U = upper result); solve $Ly = b$, then $Ux = y$ — two cheap triangular passes instead of re-eliminating. **Worked:** $A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 2 & 4 \end{pmatrix}$: $R_2 - \frac{1}{2}R_1$, $R_3 - \frac{4}{5}R'_2 \rightarrow U$ diag (2, 2.5, 3.2), det = $2 \cdot 2.5 \cdot 3.2 = 16$.

7 — Midterm Recap (compact)

- Taylor:** $f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n$, $R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$, $|R_n| \leq \frac{M_d n!}{(n+1)!}$.
- Odd fn: only odd powers (deg 8 $\rightarrow x^7$); index R_{2m+1} , bound $\frac{12^{2m+2}}{(2m+2)!}$.
- Maclaurin:** e^x , sin, cos converge $\forall \mathbb{R}$; $\frac{1}{1-x} = \sum x^k$, $\ln(1+x)$, arctan for $|x| < 1$.
- Horner:** tableau at c gives $p(c)$ (last) + quotient q ; tableau on q gives $p'(c)$.
- Interpolation:** $\ell_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$, $p = \sum y_i \ell_i$; err = $\frac{f^{(n+1)}(\xi)}{(n+1)!} \prod (x - x_i)$ ($= 0$ at nodes).
- Vandermonde det = $\prod_{i < j} (x_j - x_i) \neq 0$ (distinct nodes); DD: $f[x_i, x_{i+1}] = \frac{\Delta f}{\Delta x}$, Newton $p_n = f[x_0] + \sum f[x_0..x_k] \prod_{j < k} (x - x_j)$.

8 — Worked Mini-Examples

Coffee regression (Quiz 6)

Given $x: 0, 1, 2, 3$; $y: 3, 5, 8, 7$ ($n+1=4$ pts). Step 1 — sums (add term by term): $\sum x_i = 0+1+2+3 = 6$; $\sum y_i = 3+5+8+7 = 23$; $\sum x_i^2 = 0+1+4+9 = 14$; $\sum x_i y_i = 0+5+16+21 = 42$. Step 2 — means (divide by 4): $\bar{x} = 6/4 = 1.5$; $\bar{y} = 23/4 = 5.75$; $\bar{x}^2 = 14/4 = 3.5$; $\bar{xy} = 42/4 = 10.5$. Step 3 — coeffs (plug into boxed formula): $a_1 = \frac{10.5 - 1.5 \cdot 5.75}{3.5 - 1.5^2} = \frac{10.5 - 8.625}{3.5 - 2.25} = \frac{1.875}{1.25} = 1.5$; $a_0 = 5.75 - 1.5 \cdot 1.5 = 5.75 - 2.25 = 3.5$. $\hat{y} = 3.5 + 1.5x$. Step 4 — predict $x=4$: $\hat{y} = 3.5 + 1.5 \cdot 4 = 3.5 + 6 = 9.5$ (mark (4, 9.5) on the scatter plot with the line). **Plot recipe (Quiz6 Q2):** scatter the 4 data points, draw the line through (0, 3.5) and (4, 9.5) — regression line need NOT pass through data points. Step 5 — check fit: $\hat{y}: 3.5, 5.0, 6.5, 8.0$ vs $y: 3, 5, 8, 7$; residuals $-0.5, 0, 1.5, -1.0$; squared error $S = 0.25 + 0 + 2.25 + 1 = 3.5$ (this S is what least squares minimized). **Battery prediction:** at $t=70$: $0.714 \cdot 70 + 3.57 = 53.55\%$ (extrapolation —

flag as less reliable). **Hessian:** from Step 1 sums
 $(\sum x_i = 6, \sum x_i^2 = 14): S_{a_0 a_0} = 2 \cdot 4 = 8,$
 $S_{a_0 a_1} = 2 \cdot 6 = 12, S_{a_1 a_1} = 2 \cdot 14 = 28;$
 $\det H = 8 \cdot 28 - 12^2 = 224 - 144 = 80 > 0$ (+
 $8 > 0) \Rightarrow \min.$

Battery fit (Lec 14)

Given $t: 0, 10, 20, 30, 40, 50, 60$; charge
5, 10, 15, 25, 35, 40, 45 ($n+1=7$). Sums:
 $\sum t_i = 0+10+20+30+40+50+60 = 210;$
 $\sum y_i = 5+10+15+25+35+40+45 = 175; \sum t_i^2 =$
 $0+100+400+900+1600+2500+3600 = 9100;$
 $\sum t_i y_i = 0+100+300+750+1400+2000+2700 =$
7250. Means (divide by 7): $\bar{x} = 210/7 = 30;$
 $\bar{y} = 175/7 = 25; \bar{x^2} = 9100/7 = 1300;$
 $\overline{xy} = 7250/7 \approx 1035.71.$ Coeffs:
 $a_1 = \frac{1035.71-30 \cdot 25}{1300-30^2} = \frac{1035.71-750}{1300-900}$
 $= \frac{285.71}{400} \approx 0.714; a_0 = 25 - 0.714285 \cdot 30$
 $= 25 - 21.43 \approx 3.57.$ $\phi(x) = 0.714x + 3.57$.
Check fit:
 $\hat{y}: 3.57, 10.71, 17.85, 24.99, 32.13, 39.27, 46.41$ vs
 $y: 5, 10, 15, 25, 35, 40, 45$
(residuals
1.43, -0.71, -2.85, 0.01, 2.87, 0.73, -1.41).

Exp linearization (HW09)

Given $x: 1, 2, 3, 5, y: 8, 4, 2, 1$; model $y = Ae^{Bx} \Rightarrow$
transform $\tilde{x} = x, \tilde{y} = \ln y$. Step 0 — transform
each y : $\ln 8 \approx 2.07944, \ln 4 \approx 1.38629,$
 $\ln 2 \approx 0.69315, \ln 1 = 0$. So $\tilde{x}: 1, 2, 3, 5;$
 $\tilde{y}: 2.07944, 1.38629, 0.69315, 0$. Step 1 — means
($n+1=4$): $\bar{\tilde{x}} = (1+2+3+5)/4 = 11/4 = 2.75;$
 $\bar{\tilde{y}} = (2.07944+1.38629+0.69315+0)/4 =$
4.15888/4 ≈ 1.03972 (note = $\ln 64/4 = 1.5 \ln 2$);
 $\bar{\tilde{x}^2} = (1+4+9+25)/4 = 39/4 = 9.75;$
 $\overline{\tilde{x}\tilde{y}} = (1 \cdot 2.07944 + 2 \cdot 1.38629 + 3 \cdot 0.69315 + 5 \cdot 0)/4 =$
6.93147/4 ≈ 1.73287 (note = $2.5 \ln 2$). Step 2 —
fit: $a_1 = \frac{1.73287 - 2.75 \cdot 1.03972}{9.75 - 2.75^2} = \frac{1.73287 - 2.85923}{9.75 - 7.5625}$
 $= \frac{-1.12636}{2.1875} \approx -0.5149;$
 $a_0 = 1.03972 - (-0.5149) \cdot 2.75$
 $= 1.03972 + 1.41598 \approx 2.4557$. Step 3 —
back-transform ($A = e^{a_0}, B = a_1$):
 $A = e^{2.4557} \approx 11.65, B \approx -0.515 < 0$. Check:
 $\hat{y} = 11.65e^{-0.515x}$ gives 7.0, 4.2, 2.5, 0.9 vs 8, 4, 2, 1
— reasonable. **Nonlinear example (Lec15):**
 $x: 1, 2, 3, 4, y: 0.45, 0.81, 1.05, 1.13$ for $y = \frac{Ax}{B+x}$:
 $\tilde{x} = 1/x, \tilde{y} = 1/y$ give $a_1 \approx 1.834, a_0 \approx 0.368,$
so $A = 1/a_0 \approx 2.71, B = a_1/a_0 \approx 4.98$.

Trapezoid sin x (Quiz 7)

Given $[0, 3\pi/2]$, uniform $h = \pi/4 \Rightarrow$
 $n = (3\pi/2)/(\pi/4) = 6$ subintervals, 7 nodes
 $x_i = i\pi/4$. Step 1 — evaluate $f(x) = \sin x$:
 $\sin 0 = 0; \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} \approx 0.70711; \sin \frac{\pi}{2} = 1;$
 $\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}; \sin \pi = 0; \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2};$
 $\sin \frac{3\pi}{2} = -1$. Step 2 — uniform trapezoid
(endpoints half-weighted):
 $\tilde{I} = \frac{h}{2}(f_0+f_6) + h \sum_{i=1}^5 f_i$ with
 $\sum_{i=1}^5 f_i = \frac{\sqrt{2}}{2} + 1 + \frac{\sqrt{2}}{2} + 0 - \frac{\sqrt{2}}{2} = 1 + \frac{\sqrt{2}}{2} \approx 1.70711$.
Step 3 — multiply out: $\tilde{I} = \frac{h}{2}(f_0+f_6) + h \sum_{i=1}^5 f_i$
 $= \frac{\pi}{4} \left(\frac{1}{2}(-1) + 1.70711 \right) = \frac{\pi}{4} (1.20711) \approx$
0.78540 $\cdot 1.20711 \approx 0.94806$. Step 4 — exact
 $I = \int_0^{3\pi/2} \sin x \, dx = [-\cos x]_0^{3\pi/2} =$
 $-\cos \frac{3\pi}{2} + \cos 0 = 0 + 1 = 1$; err
 $= |1 - 0.94806| \approx 0.05194$. **How to draw**
trapezoids (Pract.6): (1) mark all $n+1$ nodes on
the axis; (2) plot $(x_i, f(x_i))$; (3) join consecutive
points with straight segments; (4) each strip is a
trapezoid (drop verticals to the axis). Parabolas
(Simpson): same nodes, one arc per pair through
its 3 points.

Simpson $\int_2^4 \frac{dx}{\ln x}$ (Quiz 8)

Given $n = 8$ (even $\checkmark$), 9 points,
 $h = (4 - 2)/8 = 0.25$. Nodes $x_i = 2 + 0.25i$:
2, 2.25, 2.5, 2.75, 3, 3.25, 3.5, 3.75, 4. Step 1 —
evaluate $f(x) = 1/\ln x$ at each node:
 $f_0 = 1/\ln 2 \approx 1.44270; f_1 = 1/\ln 2.25 \approx 1.23315;$
 $f_2 = 1/\ln 2.5 \approx 1.09136; f_3 = 1/\ln 2.75 \approx 0.98853;$
 $f_4 = 1/\ln 3 \approx 0.91024; f_5 = 1/\ln 3.25 \approx 0.84842;$

$f_6 = 1/\ln 3.5 \approx 0.79824; f_7 = 1/\ln 3.75 \approx 0.75657;$
 $f_8 = 1/\ln 4 \approx 0.72135$. Step 2 — weight (ends $\times 1,$
odd $i \times 4$, even $i \times 2$): $4 \sum_{\text{odd}} =$
 $4(1.23315+0.98853+0.84842+0.75657) =$
 $4(3.82667) = 15.30668;$
 $2 \sum_{\text{even}} = 2(1.09136+0.91024+0.79824) =$
 $2(2.79984) = 5.59968$. Step 3 — assemble:
 $I = \frac{0.25}{3}(1.44270 + 15.30668 + 5.59968 + 0.72135)$
 $= \frac{0.25}{3}(23.07041) \approx 0.83333 \cdot 23.07041 \approx 1.92253$.

Euler + midpoint $y' = x - 4y, y(0) = -3/4$ (Quiz 9)

$h = 0.2, f(x, y) = x - 4y$. **Euler:**
 $f_0 = 0 - 4(-0.75) = 3;$
 $y_1 = -0.75 + 0.2(3) = -0.15;$
 $f_1 = 0.2 - 4(-0.15) = 0.8;$
 $y_2 = -0.15 + 0.2(0.8) = 0.01;$
 $f_2 = 0.4 - 0.04 = 0.36;$
 $y_3 = 0.01 + 0.2(0.36) = 0.082$. **Midpoint:**
half-step $(0.1, -0.75 + 0.1(3)) = (0.1, -0.45),$
 $f = 0.1 + 1.8 = 1.9$
 $\rightarrow y_1 = -0.75 + 0.2(1.9) = -0.37$; half-step
 $(0.3, -0.37 + 0.1(1.68)) = (0.3, -0.202),$
 $f = 0.3 + 0.808 = 1.108$
 $\rightarrow y_2 = -0.37 + 0.2(1.108) = -0.1484$. Exact
 $y(0.4) \approx -0.101304$ (given in quiz; from closed
form $y = x/4 - 1/16 - \frac{11}{16}e^{-4x}$): Euler err
 $= |-0.101304 - 0.01| \approx 0.11130$; midpoint err
 $= |-0.101304 - (-0.1484)| \approx 0.04710$. **IVP from**
general sol: $y = -\frac{1}{4} + \frac{x}{2} + Ce^{-2x},$
 $(x_0, y_0) = (\frac{1}{2}, \frac{1}{4}): \frac{1}{4} = -\frac{1}{4} + \frac{1}{4} + Ce^{-1} \Rightarrow$
 $C = e/4 \approx 0.6796, y' = \frac{y}{x} + 2x^2, (1, 4)$, general
Q2 pattern: $y' = \frac{y}{x} + 2x^2, (1, 4)$, general
 $y = (x^2 + C)x: 4 = (\frac{1}{x} + C) \cdot 1 \Rightarrow C = 3,$
 $y^* = (x^2 + 3)x$. Euler pattern: $x_1 = 1 + h,$
 $y_1 = 4 + h(4/1 + 2) = 4 + 6h$ (since $f(1, 4) = 6$).
Midpoint pattern: $x_{1/2} = 1 + h/2,$
 $y_{1/2} = 4 + (h/2) \cdot 6 = 4 + 3h$. **Check trap:**
 $y = Ce^x - 2 \cos 2x - 2 \sin 2x$ is *not* a sol of
 $y' = y + 5 \sin 2x: y' = Ce^x + 4 \sin 2x - 4 \cos 2x,$
but $y + 5 \sin 2x = Ce^x - 2 \cos 2x + 3 \sin 2x;$
difference = $\sin 2x - 2 \cos 2x \neq 0$.

But $y = \frac{1}{2}(\cos x - \sin x) + Ce^x$ *is* a sol of
 $y' = y - \cos x:$
 $y' = \frac{1}{2}(-\sin x - \cos x) + Ce^x = y - \cos x$.
Rule: differentiate candidate AND plug into ODE
— both must hold $\forall C$. **Decision boundary**
mini-example: plane $-1 + x_1 + x_2 = 0$: point
(0, 0) gives $-1 < 0$ (class 1), (1, 1) gives $+1 > 0$
(class 2); $\sigma(-1 + x_1 + x_2)$ maps these to ≈ 0.27 and
 ≈ 0.73 . **RK4 schemes (no numbers needed):**
for $y' = y + x: k_1 = y_i + x_i, k_2 = y_i + \frac{h}{2}k_1 + x_i + \frac{h}{2},$
 $k_3 = y_i + \frac{h}{2}k_2 + x_i + \frac{h}{2}, k_4 = y_i + hk_3 + x_i + h.$
For $y' = y + 1/x$ ($h=1/12$): $k_1 = x_i y_i + 1/x_i;$
 $k_2 = (x_i + \frac{1}{24})(y_i + \frac{k_1}{24}) + 1/(x_i + \frac{1}{24}); k_3$ same with
 $k_2; k_4 = (x_i + \frac{1}{12})(y_i + \frac{k_3}{12}) + 1/(x_i + \frac{1}{12}).$

Rectangle lattice nodes (Quiz 7)

Given $x_k = \frac{3 \ln(1+k/5)}{\ln 2} - 1, k = 0..5$. Sample:
 $x_1 = \frac{3 \ln 1.2}{\ln 2} - 1 \approx -0.21090$ (since
 $3 \ln 1.2 \approx 0.54696, \ln 2 \approx 0.69315$);
 $x_0 = \frac{3 \ln 1}{\ln 2} - 1 = -1; x_5 = \frac{3 \ln 2}{\ln 2} - 1 = 2$.
Full list: -1.00000, -0.21090, 0.45628, 1.03422,
1.54399, 2.00000. $x_i^* = 0.2x_{i-1} + 0.8x_i$: -0.36872,
0.32285, 0.91863, 1.44204, 1.90880.
Rect i has width $h_i = x_i - x_{i-1}$, height $f(x_i^*)$
($f = x^3 - 1.25x^2$). **How to draw (Pract.5):** (1)
mark the $n+1$ nodes on the x -axis; (2) for each
subinterval compute x_i^* via the α -formula (any α :
 $1/3, 2/3, 0.2$, midpoint); (3) go up to the curve at
 $f(x_i^*)$ — that height is the bar top; (4) draw the
bar over $[x_{i-1}, x_i]$ (works below the axis too:
height negative $\Rightarrow$ signed area).
Accuracy table $\int_0^1 x^2 dx$ (Lec17, $h = 0.25$)
Nodes $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1; f(x) = x^2$ gives $0, \frac{1}{16}, \frac{1}{4}, \frac{9}{16}, 1$.
Left = $\frac{1}{64}$? **No:** $f_3 = (\frac{3}{4})^2 = \frac{9}{16}$, not $\frac{3}{4}$. Correct
left uses $f_0..f_3: = \frac{1}{4}(0 + \frac{1}{16} + \frac{1}{4} + \frac{9}{16}) = \frac{1}{4}(\frac{14}{16})$
 $= \frac{7}{32} \approx 0.21875$, err
 $= |0.33333 - 0.21875| \approx 0.11458$.
Mid: midpoints $\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}, f^* : \frac{1}{64}, \frac{9}{64}, \frac{25}{64}, \frac{49}{64};$

$\tilde{I} = \frac{1}{4} \cdot \frac{84}{64} \approx 0.32813$ (err 0.00521).
Right = $\frac{1}{4}(\frac{1}{16} + \frac{1}{4} + \frac{9}{16} + 1) = \frac{15}{32} \approx 0.46875$ (err
0.13542).
Trap = $\frac{11}{32} \approx 0.34375$ (err 0.01042). **Derivation**
idea (Th.10): Taylor
 $f(x) = f(x_{i-1}) + f'(\xi_x)(x - x_{i-1})$; integrate over
 $[x_{i-1}, x_i]$: local err $|E_i| \leq M_1 h_i^2/2$; sum:
 $|E| \leq \frac{M_1}{2} \sum h_i^2$; uniform: $\sum h_i^2 = nh^2,$
 $nh = b - a$
 $\Rightarrow |E| \leq \frac{M_1(b-a)}{2} h = O(h)$. Midpoint: expand at
 c_i ; odd term $\int (x - c_i) = 0$ vanishes $\Rightarrow O(h^2)$.

Simpson $\int_1^2 e^{x^2} dx$ (Lec19, $n = 4$)

Given $h = (2 - 1)/4 = 0.25$, nodes $x_i = 1 + 0.25i$:
1, $\frac{5}{4}, \frac{3}{2}, \frac{7}{4}, 2$. Step 1 — evaluate $f(x) = e^{x^2}$ (square
 x first, then exponentiate): $f_0 = e^1 \approx 2.71828;$
 $f_1 = e^{(5/4)^2} = e^{1.5625} \approx 4.77073;$
 $f_2 = e^{(3/2)^2} = e^{2.25} \approx 9.48774;$
 $f_3 = e^{(7/4)^2} = e^{3.0625} \approx 21.38094;$
 $f_4 = e^4 \approx 54.59815$.
 $I = \frac{0.25}{3}(2.71828 + 4(26.15167) + 2(9.48774) +$
54.59815) ≈ 15.0749 .
Trap check: two double-intervals $T_1 = h(f_0+f_2) =$
 $0.25(2.71828 + 9.48774) = 0.25(12.20602) \approx 3.0515,$
 $T_2 = h(f_2+f_4) = 0.25(9.48774 + 54.59815) =$
 $0.25(64.08589) \approx 16.0215$, total $\approx 19.0730 \rightarrow 19.07$
— same ballpark $\Rightarrow$ likely right. **Simpson sketch**
(Quiz8 Q1): $[0, 240]$, 6 subintervals $\Rightarrow h = 40,$
nodes $0, 40, \dots, 240$; one parabola arc per pair of
subintervals (3 arcs total); mark nodes, draw each
arc through its 3 points.

Pseudoinverse $\rightarrow$ 1D formulas (HW09)

$X^T X \bar{a} = X^T \bar{y}$ with $X = [1 \ x_i]$ gives
 $(\sum_{i=1}^{n+1} x_i \sum_{i=1}^{n+1} x_i^2) \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = (\sum_{i=1}^{n+1} y_i \sum_{i=1}^{n+1} x_i y_i)$; dividing by
 $n+1$ and solving (Cramer) returns
 $a_1 = \frac{\overline{xy} - \bar{x}\bar{y}}{\bar{x^2} - \bar{x}^2}, a_0 = \bar{y} - a_1 \bar{x}$. Cramer in full:
 $\Delta = \frac{(n+1) \sum x_i^2 - (\sum x_i)^2}{(\sum y_i)(\sum x_i^2) - (\sum x_i)(\sum x_i y_i)}$;
 $a_0 = \frac{(n+1)(\sum x_i y_i) - (\sum x_i)(\sum y_i)}{\Delta}$; dividing top
and bottom by $(n+1)^2$ gives the averaged
formulas. **HW09 Q2 setup ($m=2, n=2, 3$ pts):**
 X is 3×3 (first col of 1's, then $x_1^{(i)}, x_2^{(i)}$ rows);
 $X^+ = (X^T X)^{-1} X^T$ is 3×3 . **Q3:** $X^+ X = I$
always; $XX^+ = I$ only if X square nonsingular.

Euler $y' = y + x$ (Lec20, corrected)

$(x_0, y_0) = (0, 1), h = 0.25: y_1 = 1 + 0.25(1) = 1.25;$
 $y_2 = 1.25 + 0.25(1.5) = 1.625;$
 $y_3 = 1.625 + 0.25(2.125) = 2.15625$. (Compiled
1.5625/2.078125 used $f_1 = 1.25$ instead of
0.25 + 1.25 = 1.5 — knock-on slip.)
Exact $y^* = 2e^x - x - 1: y(0.5) \approx 1.79744,$
 $y(0.75) \approx 2.48400$ — Euler visibly below.
Midpoint on same ODE:
(0.125, 1.125) $\rightarrow y_1 = 1 + 0.25(1.25) = 1.3125$ vs
exact 1.31805 — much closer. **Separable**
pattern: $y' = x/y \Rightarrow y \, dy = x \, dx \Rightarrow y^2 = x^2 + C_1$.
Circle IVP $y' = -x/y, (2, 2): y^2/2 + x^2/2 = C = 4$
(radius 2).
IVP slip: $y' = y + x, (1, 4):$
 $4 = Ce - 2 \Rightarrow C = 6/e \approx 2.207$ (compiled 2/e
forgot +4: check $y(1) = 6 - 2 = 4 \checkmark$).

Trapezoid, non-uniform lattice

Given nodes 0, 0.25, 1 $\Rightarrow$ widths
 $h_1 = 0.25 - 0 = 0.25, h_2 = 1 - 0.25 = 0.75; f = x^2$
gives $f_0 = 0, f_1 = 0.0625, f_2 = 1$.
 $\tilde{I} = \frac{1}{2}[(f_0+f_1)h_1 + (f_1+f_2)h_2]$
 $= \frac{1}{2}[(0+0.0625)(0.25) + (0.0625+1)(0.75)]$
 $= \frac{1}{2}(0.015625 + 0.796875) = \frac{1}{2}(0.8125) = 0.40625$.
Exact $\frac{1}{3}$; err = $|0.33333 - 0.40625| \approx 0.07292$.

Laplace expansion (starred)

$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 2 & 4 \end{pmatrix}$ along row 1:
 $2(3 \cdot 4 - 1 \cdot 2) - 1(1 \cdot 4 - 1 \cdot 0) + 0 = 2 \cdot 10 - 4 = 16$
(matches elimination det = 16). Cofactor sign
 $(-1)^{i+j}$.